

SWE 4739

Embedded Software Development

Lecture 9

Composition of State Machines

Sabrina Islam

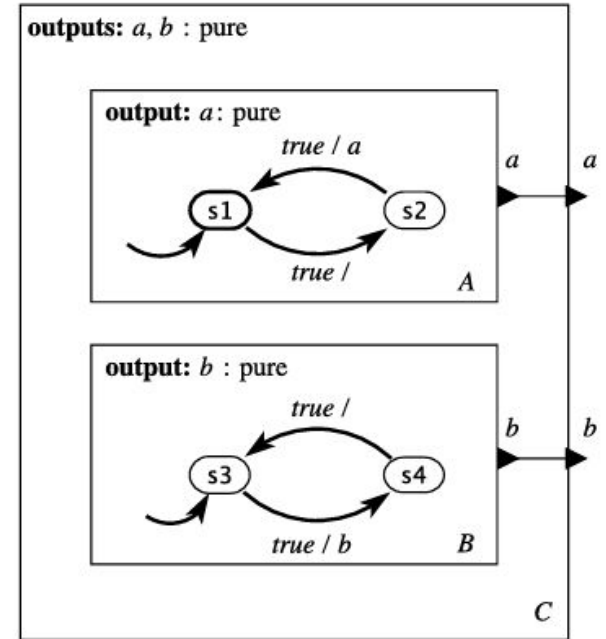
Lecturer, CSE, IUT

Contact: +8801832239897

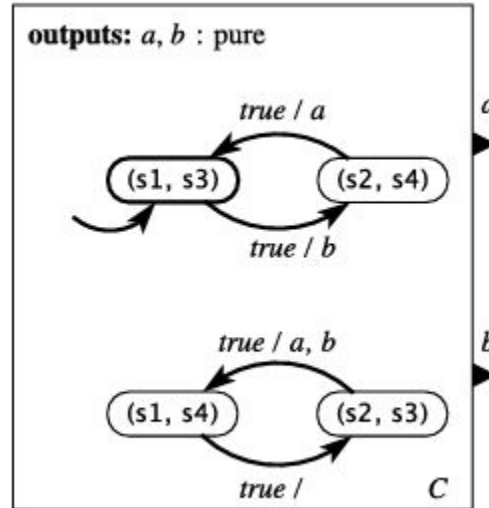
E-mail: sabrinaislam22@iut-dhaka.edu

Concurrent Composition

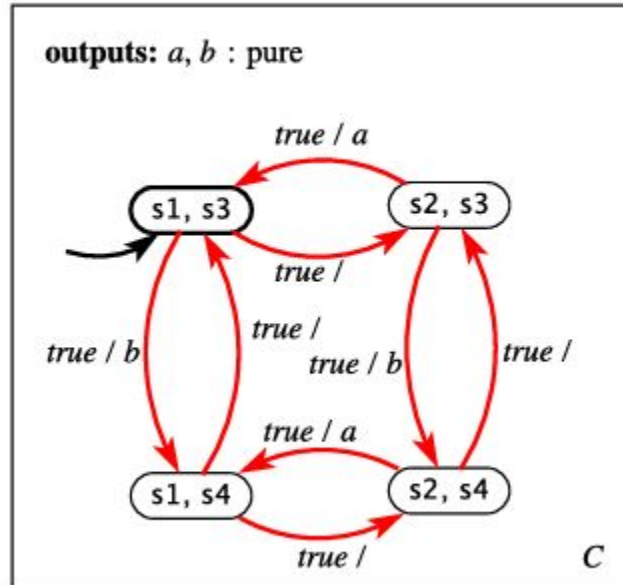
- Side-by-Side Synchronous Composition
- Side-by-Side Asynchronous Composition



Side-by-Side Synchronous Composition



Side-by-Side Asynchronous Composition

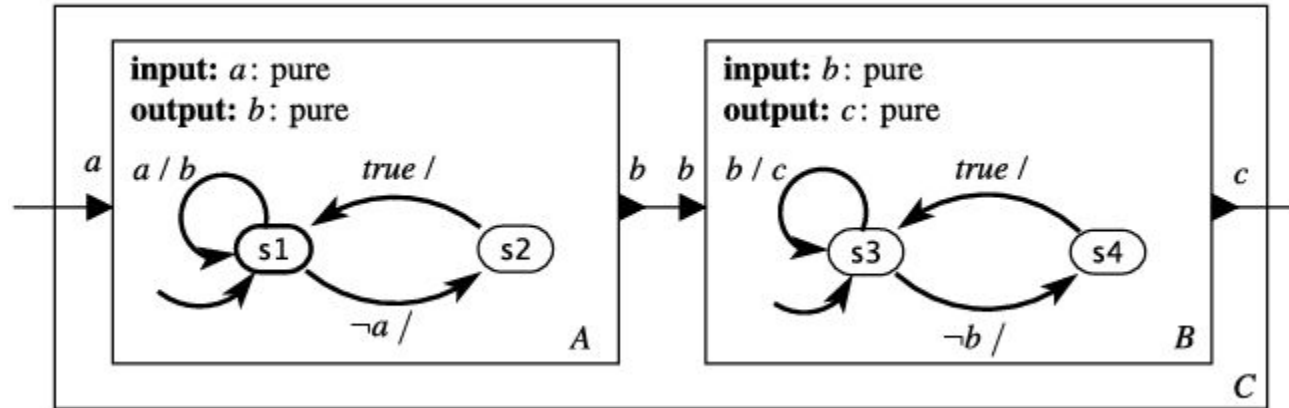


Shared Variables

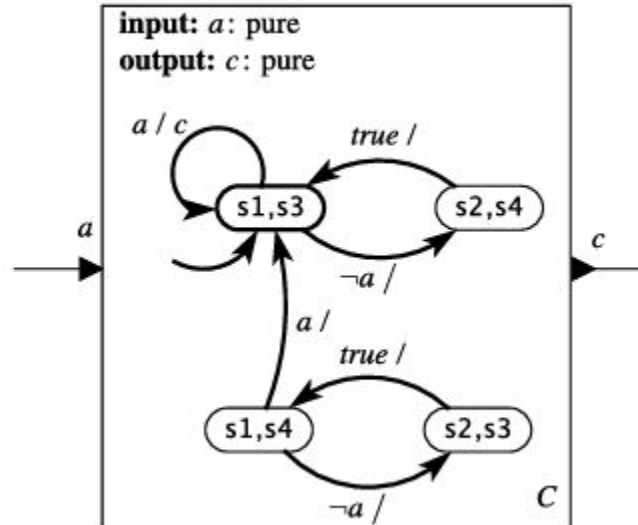
An extended state machine has local variables that can be read and written as part of taking a transition. Sometimes it is useful when composing state machines to allow these variables to be shared among a group of machines.

Example: Consider two servers that can receive requests from a network. Each request requires an unknown amount of time to service, so the servers share a queue of requests. If one server is busy, the other server can respond to a request, even if the request arrives at the network interface of the first server.

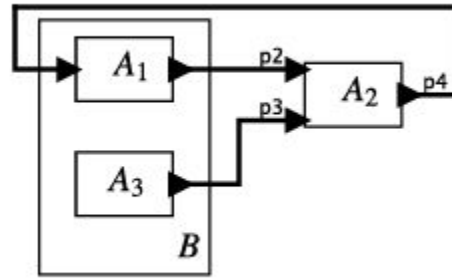
Cascade Composition



Cascade Composition



General Composition



Reference:

1. Introduction to Embedded Systems- A Cyber-Physical Systems Approach, Second Edition, E. A. Lee and S. A. Seshia.

Thank You!